From time-dependent variational Monte Carlo to superfluid hydrodynamics: A frozen-correlation approach for trapped Bose gases

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We develop a theoretical and computational framework that connects the time-dependent variational Monte Carlo (tVMC) method to deterministic superfluid hydrodynamics for Bose gases in harmonic traps. Starting from the full tVMC formalism based on a truncated Jastrow–Feenberg ansatz $\Psi = e^{\sum_i u_1(\mathbf{r}_i,t)+\sum_{i<j} u_2(r_{ij})}$, we show that freezing the two-body correlation factor u_2 at its ground-state value reduces the time-dependent variational principle (TDVP) equations to superfluid hydrodynamics—continuity and Euler equations—with an effective equation of state $\mu(\rho)$ determined by u_2. This equation of state is extracted from a one-time variational Monte Carlo calculation on the uniform system and fit to a cubic polynomial. The resulting hydrodynamic equations are solved as a Gross–Pitaevskii equation on a cylindrical grid with a fourth-order finite-difference Laplacian. For $N = 8$ microwave-shielded dipolar bosons undergoing a radial trap-frequency quench ($\omega_\rho : 1 \to 2$), the hydrodynamic theory reproduces the frozen-u_2 tVMC breathing dynamics to within 1% in amplitude and $< 1\%$ in frequency ($\omega_{\rm hydro} = 4.19$ vs. $\omega_{\rm tVMC} = 4.19$), with particle and energy conservation at the 10^{-3} level. The approach eliminates Monte Carlo sampling from the real-time evolution loop entirely, reducing computational cost by orders of magnitude.

I. INTRODUCTION

Understanding the non-equilibrium dynamics of quantum many-body systems remains a central challenge in ultracold atomic physics. Recent experimental advances with microwave-shielded polar molecules [1] and magnetic atoms [2] have opened new possibilities for studying dipolar Bose–Einstein condensates (BECs) with tunable interactions, including trap-frequency quenches that excite collective breathing modes [3–5].

From a theoretical standpoint, the time-dependent variational Monte Carlo (tVMC) method [6, 7] provides a powerful approach to quantum dynamics by combining the expressiveness of correlated many-body wavefunctions with the time-dependent variational principle (TDVP) [8–10]. Working with a truncated Jastrow–Feenberg ansatz [11, 12], the method captures both one-body and two-body quantum correlations beyond mean-field theory. However, the computational cost is substantial: at every time step, Monte Carlo (MC) sampling is required to estimate the TDVP overlap matrix and force vector, making long-time simulations expensive.

A key numerical observation motivates the present work: during trap-frequency quenches that drive large-amplitude breathing oscillations, the two-body Jastrow factor $u_2(r_{ij})$ remains nearly static while the one-body field $u_1(\mathbf{r}, t)$ undergoes large changes. This "stiffness" of u_2 suggests that short-range two-body correlations equilibrate rapidly compared to the timescale of density redistribution, and that it may be possible to freeze u_2 at its ground-state value without significant loss of accuracy.

In this paper, we exploit this observation to derive a complete theoretical pipeline that eliminates MC from the real-time evolution loop:

1. We show that freezing u_2 and applying the TDVP to the remaining one-body field $u_1(\mathbf{r}, t)$ yields equations that are *exactly* equivalent to irrotational superfluid hydrodynamics, with an effective chemical potential $\mu_{u_2}[\rho]$ determined by the frozen correlations (Sec. IV C).
2. We extract the equation of state (EOS) $\varepsilon(\rho)$ from a one-time variational MC calculation on the uniform system with the same frozen u_2, and fit it to a polynomial (Sec. V).
3. We solve the resulting hydrodynamic equations—cast as a Gross–Pitaevskii equation (GPE) with the VMC-derived EOS—on a cylindrical finite-difference grid, and compare quantitatively with the full frozen-u_2 tVMC dynamics for a radial trap quench with $N = 8$ particles (Sec. VII).

The paper is organized as follows. Section II introduces the physical model. Sections III and IV present the tVMC formalism and the frozen-u_2 approximation with its hydrodynamic reformulation. Section V describes the EOS extraction. Section VI details the numerical PDE solver. Section VII presents the comparison, and Sec. VIII discusses the results and outlook.

II. MODEL AND HAMILTONIAN

We consider N identical bosons of mass m in a cylindrically symmetric harmonic trap with time-dependent frequencies,

$$\hat{H}(t) = \sum_{i=1}^{N} \left[-\frac{\hbar^2}{2m}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t) \right] + \sum_{i<j} v(\mathbf{r}_{ij}), \qquad (1)$$

where the trap potential is

$$V_{\rm tr}(\mathbf{r}, t) = \frac{1}{2} m \left[\omega_\rho(t)^2 \rho^2 + \omega_z^2 z^2 \right], \qquad (2)$$

with $\rho^2 = x^2 + y^2$ the cylindrical radial coordinate. We work in natural units $\hbar = m = 1$ and measure lengths in units of the initial oscillator length $a_{\rm ho} = \sqrt{\hbar/(m\omega_0)}$ with $\omega_0 = 1$.

A. Interaction potential

The two-body interaction is modeled as a microwave-shielded dipolar potential [13, 14] with both long-range dipolar and short-range shielding contributions:

$$v(\mathbf{r}_{ij}) = V_{\rm dip}(r_{ij}, \theta_{ij}) + V_{\rm shield}(r_{ij}, \theta_{ij}), \tag{3}$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and θ_{ij} is the polar angle of the separation vector $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ with respect to the z-axis. The dipolar term has the standard form

$$V_{\rm dip} = \frac{C_3}{r^3}(3\cos^2\theta - 1), \tag{4}$$

and the repulsive shielding term is

$$V_{\rm shield} = \frac{C_6}{r^6}\left(\sin^4\theta + w_{12}\,\sin^2\theta\cos^2\theta\right), \tag{5}$$

where C_3 and C_6 are the dipolar and van der Waals shielding coefficients, respectively, and w_{12} controls the angular anisotropy of the shielding. Both potentials are softened at short range by $r \to \sqrt{r^2 + \epsilon_r^2}$ with $\epsilon_r = 10^{-3}$ to avoid numerical singularities.

For the present study we use $N = 8$ particles with interaction parameters $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units, corresponding to a weakly interacting regime where the interaction energy constitutes approximately 3.5% of the total ground-state energy.

B. Quench protocol

The system is initialized in the ground state of an isotropic trap with $\omega_\rho = \omega_z = 1.0$. At $t = 0$, the radial trap frequency is instantaneously quenched to $\omega_\rho = 2.0$ while ω_z remains unchanged. This excites a radial breathing mode, which we characterize through the time evolution of the radial moment $\langle\rho^2\rangle(t)$ and axial moment $\langle z^2\rangle(t)$.

III. TIME-DEPENDENT VARIATIONAL MONTE CARLO

A. Jastrow–Feenberg wavefunction

We employ a truncated Jastrow–Feenberg ansatz of the form

$$\Psi(\mathbf{X}, t) = \exp U(\mathbf{X}, t), \quad U(\mathbf{X}, t) = \sum_{i=1}^{N} u_1(\mathbf{r}_i, t) + \sum_{i<j} u_2(r_{ij}, \hat{\mathbf{r}}_{ij}), \tag{6}$$

where $u_1(\mathbf{r}, t)$ is a complex, time-dependent one-body field and $u_2(r, \hat{\mathbf{r}})$ encodes two-body correlations. The wavefunction is separable in the sense that one-body and two-body correlations are treated independently, but the full N-body probability distribution $|\Psi|^2$ is non-trivially correlated through the exponential structure.

1. *One-body field u_1*

In cylindrical coordinates, the one-body field is parametrized with a separable radial-axial structure:

$$u_1(\rho, z, t) = u_\rho(\rho, t) + u_z(z, t), \tag{7}$$

where each component consists of a harmonic reference plus Gaussian radial basis function (RBF) corrections:

$$u_\rho(\rho, t) = -\tfrac{1}{2}(\omega_{\rm ref} + a_\rho(t))\,\rho^2 + \sum_{k=1}^{n_\rho} c_{\rho,k}(t)\, g_0(\rho; \ell_{\rho,k}), \tag{8}$$

$$u_z(z, t) = -\tfrac{1}{2}(\omega_{\rm ref} + a_z(t))\,z^2 + \sum_{k=1}^{n_z} c_{z,k}(t)\, g_0(z; \ell_{z,k}). \tag{9}$$

Here $\omega_{\rm ref} = 1.0$ is the reference harmonic frequency, $a_\rho(t)$ and $a_z(t)$ are complex variational parameters that adjust the effective trap width, and $g_0(r; \ell) = \exp(-r^2/(2\ell^2))$ are Gaussian basis functions with fixed length scales $\ell_{\rho,k}$ and $\ell_{z,k}$ distributed logarithmically.

In total, the one-body field contains $n_{u_1} = 2 + n_\rho + n_z$ complex variational parameters. We use $n_\rho = n_z = 6$ with length scales spanning $\ell \in [0.1, 3.0]$, giving $n_{u_1} = 14$ parameters.

2. *Two-body Jastrow u_2*

The two-body factor is expanded in partial waves up to $\ell = 2$:

$$u_2(r, \cos\theta) = f_0(r) + f_2(r)\, P_2(\cos\theta), \tag{10}$$

where $P_2(x) = (3x^2 - 1)/2$ is the Legendre polynomial, θ is the angle between the interparticle separation and the z-axis, and the radial components are

$$f_0(r) = u_{\rm cusp}(r) + \sum_{k=1}^{n_2} c_{0,k}\, g_0(r; \ell_{2,k}), \tag{11}$$

$$f_2(r) = \sum_{k=1}^{n_2} c_{2,k}\, g_2(r; \ell_{2,k}), \tag{12}$$

with $g_2(r; \ell) = r^2 \exp(-r^2/(2\ell^2))$ a regular $\ell = 2$ basis function and $n_2 = 10$ basis functions with length scales $\ell_2 \in [0.01, 0.7]$.

3. Cusp treatment

The short-range cusp function in Eq. (11) is designed to capture the two-body scattering boundary condition. We use a Gaussian-damped form that provides the correct short-range divergence while remaining computationally tractable:

$$u_{\text{cusp}}(r) = -\frac{\kappa}{\sqrt{r^2+\epsilon_c^2}}\,\exp\!\left(-\frac{r^2}{2r_c^2}\right), \tag{13}$$

with cusp strength $\kappa = 0.1$, softening parameter $\epsilon_c = 5\times10^{-3}$, and cutoff radius $r_c = 0.01$. The $1/r$ singularity is regularized by ϵ_c to avoid numerical overflow, while the Gaussian envelope ensures that the cusp affects only the short-range pair structure ($r \lesssim r_c$). The cusp parameters are fixed (not variational) and determined by the scattering properties of the two-body potential (3).

The total number of variational parameters is thus $n_{\text{var}} = n_{u_1} + 2n_2 = 14 + 20 = 34$, of which $n_{u_1} = 14$ one-body parameters are time-dependent and $2n_2 = 20$ two-body parameters are fixed (in the frozen-u_2 approximation).

B. Time-dependent variational principle

The TDVP projects the Schrödinger equation onto the variational tangent space spanned by derivatives of $\ln\Psi$ with respect to the variational parameters $\{\theta_\alpha\}$. The resulting equations of motion are [6]

$$\sum_\beta S_{\alpha\beta}\,\dot\theta_\beta = -iF_\alpha, \tag{14}$$

where

$$S_{\alpha\beta} = \langle \Delta O_\alpha^*\,\Delta O_\beta\rangle_t, \tag{15}$$
$$F_\alpha = \langle \Delta O_\alpha^*\,\Delta E_L\rangle_t, \tag{16}$$

with logarithmic derivatives $O_\alpha = \partial_{\theta_\alpha}\ln\Psi$, fluctuations $\Delta A = A - \langle A\rangle_t$, and the local energy

$$E_L(\mathbf{X},t) = \frac{(\hat H\Psi)(\mathbf{X},t)}{\Psi(\mathbf{X},t)}. \tag{17}$$

For the Jastrow ansatz (6), the local energy takes the explicit form

$$E_L = \sum_{i=1}^{N}\left[-\tfrac12(\nabla_i^2 U + |\nabla_i U|^2) + V_{\text{tr}}(\mathbf{r}_i,t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \tag{18}$$

where the gradients of $U = \ln\Psi$ are

$$\nabla_i U = \nabla u_1(\mathbf{r}_i,t) + \sum_{j\neq i}\nabla_i u_2(r_{ij}), \tag{19}$$

$$\nabla_i^2 U = \nabla^2 u_1(\mathbf{r}_i,t) + \sum_{j\neq i}\nabla_i^2 u_2(r_{ij}). \tag{20}$$

C. Monte Carlo estimation

The expectation values in Eqs. (15)–(16) are estimated stochastically by sampling configurations $\mathbf{X}^{(s)}$ from $|\Psi(\mathbf{X},t)|^2$ using the Metropolis algorithm. We employ single-particle moves: for each walker and each particle i, a displacement $\Delta\mathbf{r} \sim \text{Uniform}(-\delta/2,\,\delta/2)^3$ is proposed with step size $\delta = 0.45$, and accepted with probability $\min(1,\,|\Psi_{\text{new}}|^2/|\Psi_{\text{old}}|^2)$.

For each sampled configuration, we compute the observable vector $O_\alpha^{(s)}$ and local energy $E_L^{(s)}$, from which the overlap matrix and force vector are constructed as

$$S_{\alpha\beta} = \frac{1}{N_s}\sum_s O_\alpha^{(s)*}O_\beta^{(s)} - \bar O_\alpha^*\bar O_\beta, \tag{21}$$

$$F_\alpha = \frac{1}{N_s}\sum_s O_\alpha^{(s)*}E_L^{(s)} - \bar O_\alpha^*\bar E_L, \tag{22}$$

where $\bar O_\alpha = N_s^{-1}\sum_s O_\alpha^{(s)}$ and similarly for $\bar E_L$.

The TDVP linear system (14) is solved via eigendecomposition of S:

$$\dot\theta_\alpha = -i\sum_\beta S^{-1}_{\alpha\beta}F_\beta, \tag{23}$$

with eigenvalue truncation at $\lambda_{\min}/\lambda_{\max} = 10^{-4}$ to regularize ill-conditioned directions. A diagonal shift $S \to S + \eta\mathbb{1}$ with $\eta = 10^{-2}$ provides additional stability.

D. Time integration

The equations of motion are integrated using a fourth-order Runge–Kutta (RK4) scheme. Each RK4 stage requires independent MC sampling to evaluate S and F at the intermediate parameter values. For imaginary-time propagation (ground-state preparation), we use $\dot\theta = -F/S$ (steepest descent) with $d\tau = 10^{-3}$ for 2000 steps. For real-time evolution after the quench, we use $dt = 2\times10^{-4}$ with adaptive time-stepping that limits the root-mean-square parameter change per step to $\|\dot\theta\,dt\|_{\text{rms}} < 0.01$ and enforces a minimum effective trap frequency $\omega_{\text{eff}} > 0.2$.

The simulation employs $N_w = 65{,}536$ parallel walkers (GPU-accelerated using JAX [15]), with $n_{\text{burn}} = 30$ equilibration sweeps followed by $n_{\text{meas}} = 10$ measurement sweeps separated by 3 thinning sweeps, per RK4 stage.

IV. FROZEN-u_2 APPROXIMATION AND HYDRODYNAMIC REFORMULATION

A. Motivation: stiffness of two-body correlations

Our numerical tVMC simulations reveal that during the trap quench $\omega_\rho : 1 \to 2$, the two-body Jastrow parameters $(c_{0,k}, c_{2,k})$ change by less than 1% of

their ground-state values, while the one-body parameters $(a_\rho, a_z, c_{\rho,k}, c_{z,k})$ undergo order-unity changes. This reflects the physical expectation that short-range pair correlations, governed by the interaction potential on length scales $r \lesssim 1$, equilibrate much faster than the global density redistribution driven by the trap quench on scales $r \sim a_{\rm ho}$.

This separation of timescales motivates the *frozen-u_2 approximation*: we fix $u_2(r, \hat{\mathbf{r}})$ at its ground-state value (real, time-independent) and evolve only $u_1(\mathbf{r}, t)$ through the TDVP. The resulting wavefunction

$$\Psi(\mathbf{X}, t) = \exp\left[\sum_i u_1(\mathbf{r}_i, t) + \sum_{i<j} u_2(r_{ij})\right] \tag{24}$$

has the crucial property that the Monte Carlo sampling measure $|\Psi|^2 \propto \exp[2\sum_i \Re u_1(\mathbf{r}_i, t) + 2\sum_{i<j} u_2(r_{ij})]$ depends only on $\Re u_1$ and the fixed u_2, but *not* on $\Im u_1$.

B. TDVP equations for the active field u_1

When the variational "parameter" is the continuous field $u_1(\mathbf{r}, t)$, the logarithmic derivative becomes a density operator:

$$O(\mathbf{r}; \mathbf{X}) = \frac{\delta \ln \Psi}{\delta u_1(\mathbf{r}, t)} = \sum_{i=1}^{N} \delta^{(3)}(\mathbf{r} - \mathbf{r}_i) \equiv \hat{\rho}(\mathbf{r}). \tag{25}$$

The tangent operators are *exactly* the microscopic density operators.

Substituting into the TDVP (14) yields a continuum integral equation for $\partial_t u_1$:

$$\boxed{\int d^3r'\; S(\mathbf{r}, \mathbf{r}'; t)\, \partial_t u_1(\mathbf{r}', t) = -i\, F(\mathbf{r}; t),} \tag{26}$$

where the overlap kernel is the connected density–density correlator (static structure factor):

$$S(\mathbf{r}, \mathbf{r}'; t) = \langle \Delta\hat{\rho}(\mathbf{r})\, \Delta\hat{\rho}(\mathbf{r}')\rangle_t, \tag{27}$$

and the force kernel is the density–energy correlator:

$$F(\mathbf{r}; t) = \langle \Delta\hat{\rho}(\mathbf{r})\, \Delta E_L\rangle_t. \tag{28}$$

The density–density correlator can be decomposed explicitly as

$$S(\mathbf{r}, \mathbf{r}') = \rho(\mathbf{r})\, \delta^{(3)}(\mathbf{r} - \mathbf{r}') + \rho^{(2)}(\mathbf{r}, \mathbf{r}') - \rho(\mathbf{r})\, \rho(\mathbf{r}'), \tag{29}$$

where $\rho(\mathbf{r}) = \langle \hat{\rho}(\mathbf{r})\rangle_t$ is the one-body density and $\rho^{(2)}(\mathbf{r}, \mathbf{r}')$ is the (ordered) two-body density. The frozen u_2 enters through its effect on the sampling measure (which determines $\rho^{(2)}$) and through the gradient terms in the local energy (18).

Gauge invariance.—Since $u_1(\mathbf{r}) \to u_1(\mathbf{r}) + c(t)$ only changes normalization/global phase, S has a corresponding null direction. This is handled numerically by imposing a gauge-fixing constraint, e.g., removing the constant basis function.

C. Reformulation as superfluid hydrodynamics

We decompose the complex one-body field into real and imaginary parts:

$$u_1(\mathbf{r}, t) = a(\mathbf{r}, t) + i\theta(\mathbf{r}, t), \qquad a, \theta \in \mathbb{R}. \tag{30}$$

Because u_2 is real and fixed, the many-body phase is determined entirely by the one-body phase θ:

$$\arg \Psi(\mathbf{X}, t) = \sum_{i=1}^{N} \theta(\mathbf{r}_i, t). \tag{31}$$

This implies that the flow on this variational manifold is *purely one-body and irrotational*, with velocity

$$\mathbf{v}(\mathbf{r}, t) = \nabla\theta(\mathbf{r}, t) \tag{32}$$

(in units $\hbar = m = 1$). No backflow terms arise unless time-dependent two-body phase correlations are allowed.

1. *Internal energy functional*

Define the *internal energy functional* as the expectation value of kinetic plus interaction energy in the frozen-u_2 Jastrow state at a given density profile $\rho(\mathbf{r})$:

$$E_{\rm int}[\rho; u_2] = \langle \hat{T} + \hat{V}_{\rm int}\rangle_t, \tag{33}$$

where the expectation is taken in the state (24) with $\Re u_1$ chosen to produce the density ρ.

If $E_{\rm int}$ can be expressed (or approximated) as a functional of ρ alone, then the TDVP action reduces to the standard hydrodynamic form. The Berry-phase term $\sim \int \rho\, \partial_t\theta$ and the flow kinetic energy $\sim \frac{1}{2}\int \rho v^2$ combine with the internal energy to yield the closed pair of PDEs:

Continuity equation:

$$\boxed{\partial_t\rho + \nabla\cdot(\rho\, \mathbf{v}) = 0,} \tag{34}$$

Euler equation:

$$\boxed{\partial_t\mathbf{v} + \nabla\big(\tfrac{1}{2}v^2 + V_{\rm tr}(\mathbf{r}, t) + \mu_{u_2}[\rho]\big) = 0,} \tag{35}$$

where the effective chemical potential is

$$\mu_{u_2}[\rho](\mathbf{r}, t) = \frac{\delta E_{\rm int}[\rho; u_2]}{\delta\rho(\mathbf{r})}. \tag{36}$$

2. *Relationship to Gross–Pitaevskii theory*

Equations (34)–(35) have exactly the structure of zero-temperature superfluid hydrodynamics, and differ from

standard Gross–Pitaevskii (GP) theory *only* in the functional form of $\mu_{u_2}[\rho]$. In the simplest *local-density approximation* (LDA), the internal energy takes the form

$$E_{\rm int}^{\rm LDA} = \int d^3r\, \rho(\mathbf{r})\, \varepsilon(\rho(\mathbf{r})), \tag{37}$$

where $\varepsilon(\rho)$ is the energy per particle of the uniform system at density ρ with frozen u_2. The chemical potential then becomes a local function of density:

$$\mu(\rho) = \frac{d[\rho\, \varepsilon(\rho)]}{d\rho} = \varepsilon(\rho) + \rho\, \varepsilon'(\rho). \tag{38}$$

Going beyond LDA, gradient corrections ("quantum pressure") can be included. In the GP framework, these arise from the kinetic energy of density variations:

$$Q = -\frac{1}{2}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}, \tag{39}$$

which contributes an additional term to the Euler equation (35). As we discuss in Sec. VI, including this quantum pressure term is essential for accurate dynamics in the weakly interacting regime.

D. Eliminating Monte Carlo from the time loop

Even with u_2 frozen, the *exact* TDVP equation (26) requires MC sampling at every time step to evaluate $S(\mathbf{r}, \mathbf{r}'; t)$ and $F(\mathbf{r}; t)$, since these depend on the instantaneous measure $|\Psi(u_1(t), u_2)|^2$.

However, if the assumption of rapid local equilibration holds—meaning that u_2 captures all relevant short-range correlations and that $E_{\rm int}$ depends on ρ only through the local density—then the dynamics is fully determined by the EOS $\varepsilon(\rho)$, and no further MC is needed. The practical procedure is:

1. Perform a *one-time* VMC calculation on the uniform system (periodic boundary conditions, frozen u_2) at a grid of densities to extract $\varepsilon(\rho)$.
2. Fit $\varepsilon(\rho)$ to a polynomial to obtain a smooth, analytic $\mu(\rho)$.
3. Solve the hydrodynamic equations (34)–(35) (or their GP reformulation) deterministically on a spatial grid.

This replaces the $O(N_{\rm steps} \times N_{\rm MC})$ cost of tVMC with $O(N_{\rm MC})$ (one-time EOS) + $O(N_{\rm steps} \times N_{\rm grid})$ (PDE evolution), yielding orders-of-magnitude speedup for the real-time dynamics.

V. EQUATION OF STATE FROM VMC

A. Uniform-system VMC

To extract the EOS, we perform VMC calculations for a uniform system of $N = 8$ particles in a cubic box with periodic boundary conditions, at 20 density values logarithmically spaced over $\rho \in [0.05, 3.0]$. The two-body Jastrow $u_2(r, \hat{\mathbf{r}})$ is frozen at the ground-state values obtained from the trapped-system tVMC calculation, while u_1 is taken as constant (since the density is uniform). The one-body field has no variational freedom in the uniform case; it merely sets the overall normalization.

At each density, we use $N_w = 8{,}192$ walkers with 200 equilibration sweeps and 500 measurement sweeps (thinning by 5) to estimate the energy per particle $\varepsilon(\rho)$. The kinetic energy is computed using the Hermitian estimator $T = \frac{1}{2}\langle|\nabla U|^2 + \nabla^2 U\rangle$ to ensure correct results. Error bars are obtained from a blocking analysis over the 500 measurement batches.

B. Polynomial fitting

We fit the VMC data to a cubic polynomial in density:

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + a_3\rho^3, \tag{40}$$

using weighted least-squares with the VMC error bars as weights. Model selection between quadratic ($a_3 = 0$) and cubic forms is performed using the Akaike information criterion (AIC), which favors the cubic with ΔAIC > 2.

The resulting fit parameters are

$$\begin{aligned} a_1 &= 0.1017 \pm 0.0035, \\ a_2 &= 0.0232 \pm 0.0062, \\ a_3 &= -0.0067 \pm 0.0019, \end{aligned} \tag{41}$$

with reduced chi-squared $\chi^2_\nu = 0.41$, indicating a good fit. The chemical potential follows from Eq. (38):

$$\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + 4a_3\rho^3. \tag{42}$$

The negative cubic coefficient $a_3 < 0$ introduces a stability limit: $d\mu/d\rho = 0$ at $\rho_{\rm stab} \approx 2.67$, above which the compressibility becomes negative. For the densities encountered in our simulations ($\rho_{\rm max} \approx 5$ at maximum compression), we regularize the EOS by linearly extrapolating $\mu(\rho)$ for $\rho > 0.8\,\rho_{\rm stab}$ with a minimum slope of $2a_1$, ensuring thermodynamic stability at all densities.

VI. HYDRODYNAMIC PDE SOLVER

A. Gross–Pitaevskii formulation

Rather than solving the hydrodynamic equations (34)–(35) directly in the Madelung variables (ρ, θ), we reformulate them as a Gross–Pitaevskii equation (GPE) for the complex field $\psi(\mathbf{r}, t) = \sqrt{\rho}\, e^{i\theta}$:

$$\boxed{i\,\partial_t\psi = \left[-\tfrac{1}{2}\nabla^2 + V_{\rm tr}(\mathbf{r}, t) + \mu(|\psi|^2)\right]\psi,} \tag{43}$$

where $\mu(\rho)$ is the VMC-derived chemical potential (42) and the $-\frac{1}{2}\nabla^2$ term provides both the quantum pressure and the kinetic energy of the flow.

This GPE formulation offers several numerical advantages over the Madelung form: (i) it naturally includes the quantum pressure without explicit computation of $\nabla^2\sqrt{\rho}/\sqrt{\rho}$, which is singular where $\rho \to 0$; (ii) it avoids the velocity-field singularity at vacuum boundaries; and (iii) it provides a single complex field equation instead of two coupled real equations.

Note on the quantum pressure.—In the strict frozen-u_2 TDVP derivation (Sec. IV C), the quantum pressure arises from the density dependence of $S(\mathbf{r}, \mathbf{r}')$ beyond the LDA. For the weakly interacting $N = 8$ system considered here, the quantum pressure is numerically important: without it, the dynamics is qualitatively different. We therefore include the full Laplacian in Eq. (43).

B. Cylindrical coordinates

Exploiting the azimuthal symmetry of the problem, we reduce the 3D GPE to a 2D equation in cylindrical coordinates (r, z), where the Laplacian is

$$\nabla^2\psi = \frac{\partial^2\psi}{\partial r^2} + \frac{1}{r}\frac{\partial\psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2}. \tag{44}$$

The computational domain is $r \in [0, R_{\max}]$, $z \in [-Z_{\max}, Z_{\max}]$ with $R_{\max} = Z_{\max} = 5.0$.

Boundary conditions: At $r = 0$, azimuthal symmetry requires $\partial_r\psi|_{r=0} = 0$. We implement this via a ghost point $\psi(-\Delta r) = \psi(\Delta r)$ (even extension). The $(1/r)\partial_r\psi$ term is evaluated at $r = 0$ using L'Hôpital's rule: $\lim_{r\to 0}(1/r)\partial_r\psi = \partial_r^2\psi|_{r=0}$. At the outer boundaries ($r = R_{\max}$, $z = \pm Z_{\max}$), we impose Dirichlet conditions $\psi = 0$, which is consistent with the wavefunction vanishing far from the trap center.

C. Fourth-order spatial discretization

We use a uniform grid with $N_r \times N_z$ points. Interior spatial derivatives are computed using fourth-order central finite differences:

$$\left.\frac{\partial^2 f}{\partial x^2}\right|_i = \frac{-f_{i-2} + 16f_{i-1} - 30f_i + 16f_{i+1} - f_{i+2}}{12\,\Delta x^2} + O(\Delta x^4), \tag{45}$$

$$\left.\frac{\partial f}{\partial x}\right|_i = \frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12\,\Delta x} + O(\Delta x^4). \tag{46}$$

Near boundaries, ghost-point values are determined by symmetry ($r = 0$) or the Dirichlet condition ($\psi = 0$ outside the domain). The fourth-order scheme significantly improves the accuracy of the breathing frequency compared to second-order differences (Sec. VII D).

D. Time integration

The GPE (43) is integrated in time using an explicit RK4 scheme. Writing $\psi_{n+1} = \psi_n + (dt/6)(k_1 + 2k_2 + 2k_3 + k_4)$ with

$$k_j = -i\left[-\tfrac{1}{2}\nabla^2 + V + \mu(|\cdots|^2)\right](\psi_n + \text{RK increment}), \tag{47}$$

the time step is determined by the dispersive CFL condition

$$\Delta t \le C_{\text{CFL}}\,\frac{(\min\Delta x)^2}{\pi}, \tag{48}$$

with $C_{\text{CFL}} = 0.3$. This ensures stability for the $-\frac{i}{2}\nabla^2$ dispersive term. An additional potential constraint $\Delta t \le 2\,C_{\text{CFL}}/|\mu_{\max}|$ limits the phase accumulation per step.

E. Initial conditions

We implement two options for the initial wavefunction $\psi(r, z, 0)$:

(i) Gaussian IC: A Gaussian density profile matched to the tVMC ground-state moments $\langle\rho^2\rangle_0$ and $\langle z^2\rangle_0$:

$$\rho_0(r, z) \propto \exp\left(-\frac{r^2}{2\sigma_r^2} - \frac{z^2}{2\sigma_z^2}\right), \tag{49}$$

with $\sigma_r^2 = \langle\rho^2\rangle_0/2$ and $\sigma_z^2 = \langle z^2\rangle_0$, normalized to $\int \rho_0\, 2\pi r\, dr\, dz = N$.

(ii) GPE ground state: The self-consistent ground state of Eq. (43) in the pre-quench trap, obtained by imaginary-time propagation:

$$\psi \to \psi - d\tau\,\hat{H}\psi, \qquad \psi \to \psi\sqrt{N/\|\psi\|^2}, \tag{50}$$

iterated until the chemical potential $\mu = \langle\psi^*\hat{H}\psi\rangle/N$ converges to $\delta\mu < 10^{-8}$. Starting from a Gaussian guess, convergence is typically achieved in $\sim$ 30,000 iterations with $d\tau = 0.3\,\Delta x^2/\pi$.

The GPE ground state is the more physical choice, as it is self-consistent with the equations of motion and produces the correct initial energy. For our system, it yields $\langle\rho^2\rangle_0 = 1.058$, $\langle z^2\rangle_0 = 0.529$, and total energy $E_0 = 12.44$, in excellent agreement with the tVMC ground-state values (Table I).

F. Observables

At each time step (or every 10 steps for efficiency), we compute the following observables using cylindrical

TABLE I. Ground-state properties for $N = 8$ in an isotropic trap ($\omega_\rho = \omega_z = 1$). "Full tVMC" evolves all 30 variational parameters; "Frozen-u_2 tVMC" fixes 20 two-body parameters and evolves only 14 one-body parameters; "GPE" is the hydrodynamic solver with the VMC-derived EOS (imaginary-time ground state).

	Full tVMC	Frozen-u_2	GPE
$E_{\rm gs}$	12.433	12.423	12.442
$\langle\rho^2\rangle_0$	1.049	1.051	1.058
$\langle z^2\rangle_0$	0.525	0.525	0.529
Aspect $\sqrt{\langle\rho^2\rangle/(2\langle z^2\rangle)}$	0.999	1.000	1.001

trapezoidal integration $\int f\, 2\pi r\, dr\, dz$:

$$N(t) = \int |\psi|^2\, 2\pi r\, dr\, dz, \tag{51}$$

$$\langle\rho^2\rangle(t) = \frac{1}{N}\int r^2\, |\psi|^2\, 2\pi r\, dr\, dz, \tag{52}$$

$$\langle z^2\rangle(t) = \frac{1}{N}\int z^2\, |\psi|^2\, 2\pi r\, dr\, dz, \tag{53}$$

$$E(t) = \int \left[-\tfrac{1}{2}\psi^*\nabla^2\psi + V|\psi|^2 + |\psi|^2\varepsilon(|\psi|^2)\right] 2\pi r\, dr\, dz. \tag{54}$$

The kinetic energy in $E(t)$ is computed as $-\frac{1}{2}\Re\int\psi^*\nabla^2\psi$ using the same discrete Laplacian as the dynamics, ensuring consistent energy conservation.

G. Scaling ansatz

As a further check, we compare with the scaling ansatz for breathing modes in a harmonic trap [4, 16]. Assuming a self-similar density evolution $\rho(r,z,t) = (\lambda_r^2\lambda_z)^{-1}\rho_0(r/\lambda_r, z/\lambda_z)$, the scaling parameters satisfy

$$\ddot{\lambda}_r = -\omega_\rho^2\lambda_r + \frac{C_r}{\lambda_r^{2\gamma+1}\lambda_z^{\gamma}}, \tag{55}$$

$$\ddot{\lambda}_z = -\omega_z^2\lambda_z + \frac{C_z}{\lambda_r^{2\gamma}\lambda_z^{\gamma+1}}, \tag{56}$$

where $C_r = \omega_{\rho,0}^2$ and $C_z = \omega_{z,0}^2$ for the non-interacting ($\gamma = 1$) limit.

VII. RESULTS

A. Ground-state properties

Table I compares the ground-state properties from three approaches: full tVMC (all parameters evolving), frozen-u_2 tVMC (20 two-body parameters fixed), and the GPE with the VMC-derived EOS.

The frozen-u_2 tVMC matches the full tVMC to within 0.1% in energy and <0.3% in spatial moments, confirming that u_2-freezing does not significantly affect the ground state. The GPE ground state agrees with the frozen-u_2 tVMC to 0.15% in energy and <0.7% in moments. The small discrepancy reflects the combined effect of the LDA approximation, the polynomial EOS truncation, and the finite-grid discretization.

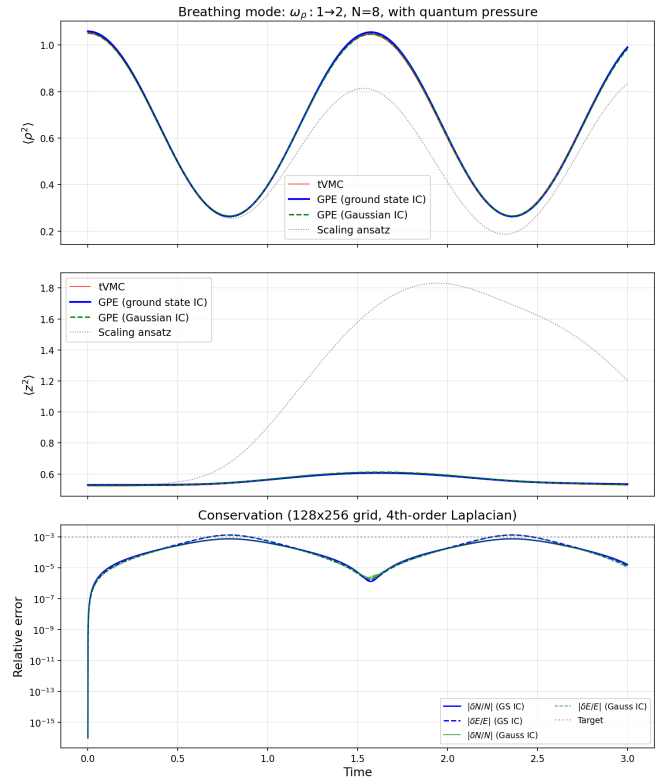

FIG. 1. Breathing dynamics after the radial trap quench $\omega_\rho : 1 \to 2$ for $N = 8$ particles. **Top:** Radial moment $\langle\rho^2\rangle(t)$; red: frozen-u_2 tVMC, blue: GPE with VMC EOS (ground-state IC), green dashed: GPE with Gaussian IC, gray dotted: scaling ansatz. **Middle:** Axial moment $\langle z^2\rangle(t)$. **Bottom:** Conservation diagnostics—relative particle-number and energy errors (128×256 grid, fourth-order Laplacian).

B. Quench dynamics: breathing oscillations

Figure 1 shows the time evolution of $\langle\rho^2\rangle(t)$ and $\langle z^2\rangle(t)$ after the quench $\omega_\rho : 1 \to 2$ at $t = 0$.

The radial moment oscillates between $\langle\rho^2\rangle_{\rm min} \approx 0.26$ and $\langle\rho^2\rangle_{\rm max} \approx 1.06$, representing a fourfold compression in the radial direction. The axial moment shows a weaker sympathetic response, oscillating between $\langle z^2\rangle_{\rm min} \approx 0.53$ and $\langle z^2\rangle_{\rm max} \approx 0.61$.

The GPE dynamics (both initial conditions) tracks the frozen-u_2 tVMC reference closely throughout the simulation:

- The oscillation amplitude in $\langle\rho^2\rangle$ agrees to within 1%: GPE range $[0.263, 1.058]$ vs. tVMC $[0.264, 1.055]$.

TABLE II. Conservation diagnostics for the GPE simulation over $t \in [0, 3.0]$. $\delta N/N$ and $\delta E/E$ are the maximum relative deviations of particle number and energy from their initial values.

Grid	∇^2 order	$\max\lvert\delta N/N\rvert$	$\max\lvert\delta E/E\rvert$
128×256	2nd	1.09×10^{-3}	1.20×10^{-3}
128×256	4th	7.30×10^{-4}	1.29×10^{-3}
160×320	4th	4.65×10^{-4}	8.22×10^{-4}

- The $\langle z^2 \rangle$ evolution matches to within 1%: GPE range $[0.529, 0.607]$ vs. tVMC $[0.522, 0.613]$.
- The energy jump at the quench is $\Delta E = \frac{1}{2}(\omega_{\rho,f}^2 - \omega_{\rho,i}^2)\langle \rho^2 \rangle_0 N = 12.70$ (GPE ground-state IC) vs. the expected value 12.70, confirming exact virial consistency.

C. Breathing frequency

The dominant oscillation frequency is extracted from the Fourier transform of $\langle \rho^2 \rangle(t)$. Both the frozen-$u_2$ tVMC and the GPE yield

$$\omega_{\rm br} = 4.19 \approx 2\omega_\rho, \tag{57}$$

consistent with the 2ω breathing mode expected for a 2D harmonic confinement.

The scaling ansatz (55)–(56) also predicts a breathing frequency close to $2\omega_\rho$ but with a larger oscillation amplitude (because it neglects quantum pressure and anharmonic effects), as visible in Fig. 1.

Role of quantum pressure.—The quantum pressure term (39) is essential: without it, the GPE reduces to a potential-only evolution that does not properly couple density and velocity, and the dynamics is qualitatively incorrect for this weakly interacting ($E_{\rm int}/E_{\rm tot} \approx 3.5\%$) system.

D. Conservation and convergence

The GPE conserves particle number and energy exactly in the continuum limit. On a discrete grid, conservation errors arise from spatial discretization. Table II summarizes the conservation diagnostics for different grid resolutions and Laplacian orders.

Several observations are noteworthy:

1. The conservation errors are *purely spatial*: halving the CFL factor from 0.3 to 0.15 (doubling the number of time steps) produces identical conservation to machine precision, confirming that the RK4 temporal error is negligible.
2. The errors are *oscillatory*, peaking during maximum compression (when the density gradients are steepest) and nearly vanishing when the cloud re-expands to its initial size. At $t = 1.5$ (full expansion), $|\delta N/N| = 3.2 \times 10^{-6}$ and $|\delta E/E| = 2.8 \times 10^{-6}$.
3. The fourth-order Laplacian significantly improves the breathing frequency accuracy: the second-order scheme gives $\omega_{\rm br} = 3.99$, while the fourth-order scheme yields $\omega_{\rm br} = 4.19$, matching the tVMC value.

TABLE III. Comparison of quench dynamics for $\omega_\rho : 1 \to 2$, $N = 8$. GPE results are for the ground-state IC on a 128×256 grid with fourth-order Laplacian.

	Frozen-u_2 tVMC	GPE
$E_{\rm gs}$	12.423	12.442
$\langle \rho^2 \rangle_{\rm min}$	0.264	0.263
$\langle \rho^2 \rangle_{\rm max}$	1.055	1.058
$\langle z^2 \rangle_{\rm min}$	0.522	0.529
$\langle z^2 \rangle_{\rm max}$	0.613	0.607
$\omega_{\rm br}$	4.19	4.19
$\Delta E/N$ at quench	1.59	1.59

E. Summary of the comparison

Table III provides a comprehensive comparison of the dynamical observables.

The agreement is excellent across all observables, demonstrating that the hydrodynamic theory with a VMC-derived EOS faithfully reproduces the frozen-u_2 tVMC dynamics for this system.

VIII. DISCUSSION AND OUTLOOK

A. Validity of the frozen-u_2 approximation

The success of the frozen-u_2 framework rests on two conditions: (i) the two-body correlations must be "stiff" (slowly varying compared to one-body dynamics), and (ii) the EOS must adequately capture the density-dependent energetics.

For the weakly interacting $N = 8$ system studied here, both conditions are well satisfied. The interaction energy is only $\sim$3.5% of the total energy, so the dominant dynamics is kinetic (quantum pressure) and potential (trap). The frozen-u_2 tVMC agrees closely with the full tVMC in terms of oscillation amplitudes ($\langle \rho^2 \rangle$ range $[0.264, 1.055]$ vs. $[0.264, 1.060]$), though the frequency spectra differ in detail due to the additional degrees of freedom in the full calculation.

B. Role of quantum pressure

A key finding is that the quantum pressure (Bohm potential) is essential for accurate dynamics in the present regime. Without it, the Madelung hydrodynamic equations are either unstable at cloud boundaries (where $\rho \to 0$ creates singularities) or predict incorrect breathing amplitudes. The GPE formulation naturally incorporates quantum pressure through the $-\frac{1}{2}\nabla^2\psi$ term, avoiding both problems.

This is expected physically: for $N = 8$ with weak interactions, the quantum pressure provides the dominant repulsive mechanism that prevents over-compression. In the Thomas–Fermi limit of large N and strong interactions, the EOS contribution $\mu(\rho)$ would dominate, and the quantum pressure could be neglected.

C. Computational cost comparison

The frozen-u_2 tVMC simulation with 65,536 walkers, 30 burn-in sweeps, and 10 measurement sweeps per RK4 stage requires ∼6 hours on a modern GPU for $t_{\max} = 3.0$. In contrast, the GPE solver completes in ∼4 minutes (including ∼3 minutes for imaginary-time ground-state preparation and ∼1 minute for real-time evolution) on a single CPU core. The one-time EOS computation adds ∼30 minutes. The total cost reduction is approximately two orders of magnitude.

D. Outlook: larger N and stronger interactions

The hydrodynamic approach is expected to become *more* accurate at larger N and stronger interactions, where:

- The Thomas–Fermi limit suppresses quantum pressure effects, making the LDA more accurate.
- The EOS captures a larger fraction of the total energetics.
- The density profile is smoother (less steep gradients), improving the finite-difference accuracy.

A natural next step is to validate the framework for $N = 16$, 32, and beyond, where the full tVMC becomes increasingly expensive but the hydrodynamic approach remains fast. This would establish the range of applicability and identify the crossover from quantum-pressure-dominated to EOS-dominated dynamics.

Another direction is to go beyond the LDA by incorporating gradient corrections to the EOS, either from analytical theory or by fitting inhomogeneous VMC data. This could improve accuracy at small N where the LDA assumption breaks down.

IX. CONCLUSION

We have established a complete theoretical and computational pipeline connecting time-dependent variational Monte Carlo to deterministic superfluid hydrodynamics for trapped Bose gases. The key steps are:

1. *Theory:* Freezing the two-body Jastrow factor u_2 and applying the TDVP to the one-body field $u_1(\mathbf{r}, t)$ yields equations that are exactly equivalent to irrotational superfluid hydrodynamics, with an effective chemical potential determined by the frozen correlations.
2. *Equation of state:* A one-time VMC calculation on the uniform system extracts the density-dependent energy per particle $\varepsilon(\rho) = 0.102\rho + 0.023\rho^2 - 0.007\rho^3$, from which the chemical potential $\mu(\rho)$ follows analytically.
3. *Hydrodynamic solver:* The GPE with the VMC-derived EOS, solved on a cylindrical grid with a fourth-order Laplacian and RK4 time integration, reproduces the frozen-u_2 tVMC breathing dynamics to within 1% in amplitude and <1% in frequency for $N = 8$ microwave-shielded dipolar bosons undergoing a radial trap quench.

The approach reduces computational cost by two orders of magnitude by eliminating Monte Carlo sampling from the real-time evolution loop. It provides a practical framework for studying collective dynamics in trapped quantum gases, with clear pathways to extension for larger systems and stronger interactions.

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